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Primary Vesicoureteral Reflux and Reflux Nephropathy

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INTRODUCTION

Vesicoureteral reflux (VUR) may be primary or secondary. Primary VUR is diagnosed mostly after a urinary tract infection (UTI) or during work-up for antenatal hydronephrosis. Secondary VUR results mostly from increased bladder pressure because of obstructive uropathy or neurogenic bladder. In this chapter, we discuss primary VUR only.

Prenatal kidney injury in the form of dysplasia has been noted with high-grade primary VUR. The presence of VUR increases the risk for postnatal upper tract infection (pyelonephritis), and the two together can cause kidney injury leading to scarring of the kidney termed *reflux nephropathy* (RN). RN may manifest as proteinuria, hypertension, preeclampsia, chronic kidney disease (CKD), and even end-stage kidney disease (ESKD). Some patients present with proteinuria as a result of secondary focal segmental glomerulosclerosis (FSGS).

CLASSIFICATION

VUR is classified by radiologic evaluation on voiding cystourethrography (VCU) into five grades as defined by the International Reflux Study in Children (Fig. 62.1 and Table 62.1).[1] An example of grade V reflux is shown in Fig. 62.2. Grades I and II are considered low-grade VUR, and grades III to V are considered high-grade VUR. Grading of VUR is necessary to standardize management strategies and compare clinical outcomes. Although widely used, there is interobserver variability in the application of the grading system.[2]

EPIDEMIOLOGY

The reported incidence of VUR depends on the age of presentation and sex. VUR is often first suggested by dilation of the fetal kidney during ultrasound examination. It is suspected when the fetal kidney pelvis is more than 5 mm in anteroposterior diameter; a diameter of more than 10 mm is suggestive of high-grade VUR. In neonates who had evidence of fetal kidney pelvis dilation, as many as 13% to 22% have VUR. Indeed, it is estimated that 1% to 2% of neonates have VUR, with a higher frequency in boys and premature infants.[3] The incidence of kidney dysplasia is also greater in male infants with VUR.[4] Although some studies show that there is a higher incidence of VUR in females compared with males, this may be attributable to the higher predisposition of UTIs in females, therefore leading to greater diagnosis of VUR in females.

VUR is identified in 30% to 40% of children presenting with UTI, predominantly in girls. VUR may also be identified in as many as 40% of siblings of index patients. When identified using sibling screening protocols, VUR was felt to be less likely to be associated with infection or long-term kidney injury.[5] VUR is less common and less severe in Black children.[6] Only about one-third as many Black girls as White girls with UTI have VUR, and no significant differences in age or mode of presentation exist between the two races.

EMBRYOLOGY AND PATHOLOGY

Primary VUR is a congenital anomaly of the ureterovesical junction caused by shortening of the intravesical submucosal length of the ureter, leading to an incompetent valve mechanism (Fig. 62.3). The formation of the ureteral bud from the mesonephric duct signals the initial development of the metanephric kidney, the final stage of kidney development. The ureteral bud interacts with the mesenchyme to give rise to the metanephric kidney. As the mesonephric duct is gradually absorbed into the enlarging urogenital sinus (the precursor of the developing bladder), the location of the ureteral bud on the mesonephric duct plays a role in the eventual location of the ureteral meatus within the bladder. If the ureteral bud reaches the urogenital sinus too early because of the absorption pattern of the mesonephric duct, it is eventually located more laterally and proximally in the bladder. This location is associated with reduction in the intravesical submucosal length of the ureter, leading to VUR.

The ureterovesical junction is designed to prevent free reflux of urine from the bladder to the kidney. The ureters pass into the bladder through the detrusor in an oblique path. The distal end of the ureter is located submucosally within the bladder. The length of the submucosal ureter is critical for preventing VUR. The muscles of the ureter extend into the trigone of the bladder and mesh with the fibers from the opposite ureter. This intermingling of fibers helps anchor the ureters into the trigone of the bladder. The distal submucosal segment is compressed against the muscular bladder wall with bladder filling, acting as an additional mechanism to prevent reflux. Because urine is propelled antegrade down the ureter, the tone of the ureter and the meatus in the bladder also help prevent reflux.

CLINICAL MANIFESTATIONS

Presentation of Primary Vesicoureteral Reflux

The three most common presentations are during follow-up for antenatal hydronephrosis, after a diagnosis of UTI, and on screening of siblings of a patient with VUR (Box 62.1).

Reflux Identified Secondary to Antenatal Hydronephrosis

Diagnosis of VUR may be suspected in utero with unilateral or bilateral hydronephrosis and confirmed after birth with VCU. There is a higher incidence of male infants diagnosed with VUR after identification of antenatal hydronephrosis.[7] Spontaneous resolution of the VUR occurs more commonly in boys with lower grades and unilateral reflux; infection rates are also lower in this cohort.[8] Female infants are more likely to have lower grades of VUR and are also less likely to develop kidney

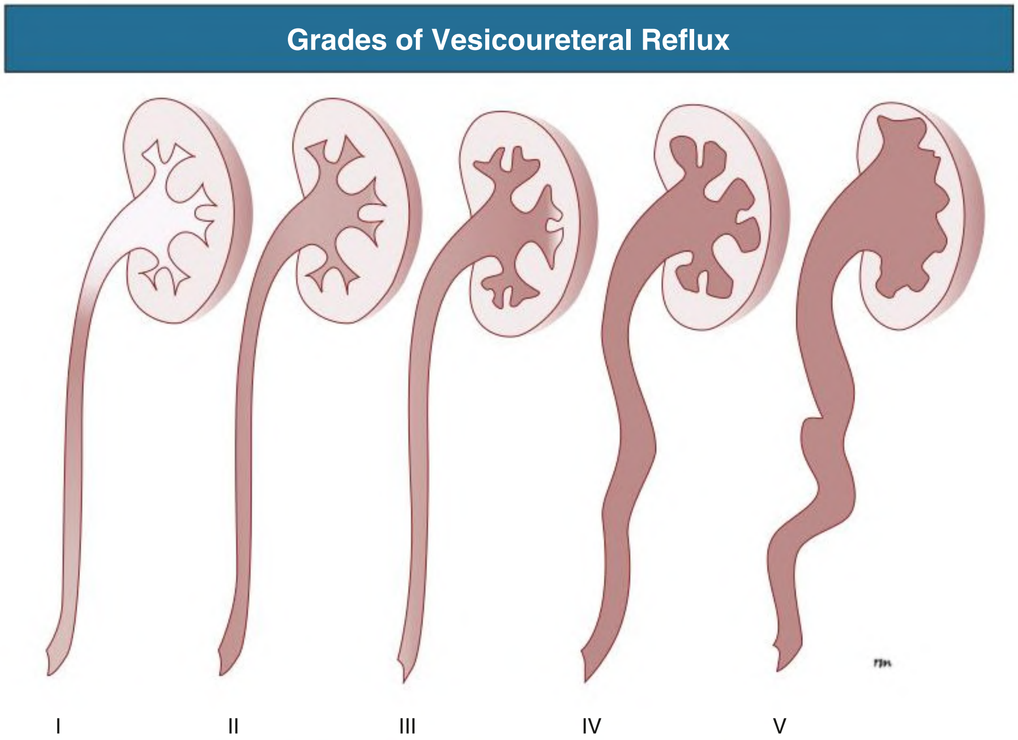

Fig. 62.1 Grades of vesicoureteral reflux.

TABLE 62.1 Classification of VUR[a]

Grade	Degree of VUR
I	Ureter only
II	Reflux into ureter, pelvis, and calyces with no dilation and with normal calyceal fornices
III	Mild or moderate dilation and/or tortuosity of the ureter and mild or moderate dilation of the pelvis; no or slight blunting of the fornices
IV	Moderate dilation and/or tortuosity of the ureter and moderate dilation of the pelvis and calyces; complete obliteration of the pelvis and calyces; complete obliteration of the sharp angles of the fornices but maintenance of the papillary impressions in the majority of the calyces (see Fig. 62.7C)
V	Gross dilation and tortuosity of the ureter, pelvis, and calyces; the papillary impressions are no longer visible in the majority of calyces (see Fig. 62.2)

[a]According to the International Reflux Study in Children.
VUR, Vesicoureteral reflux.

Fig. 62.2 **Gross Vesicoureteral Reflux (VUR) and Intrarenal Reflux.** A voiding cystourethrogram shows grade V VUR with intrarenal reflux into several kidney lobes in an infant.

damage compared with newborn males. Not all patients with antenatal hydronephrosis are investigated for the presence of VUR.

Reflux Identified After a Urinary Tract Infection

VUR is most commonly identified after UTI, particularly in a young child. The prevalence of VUR is higher in younger patients and decreases with age (Table 62.2). In neonates and toddlers, UTI may manifest as failure to thrive as opposed to typical symptoms of dysuria and frequency. VUR is more common in patients with complicated or upper tract UTI. Because VUR may potentiate the effect of UTI in children, the recommendations for evaluation of UTI have included ultrasound and VCU after resolution of symptoms. Current guidelines from the National Institute for Health and Care Excellence (NICE) and the American Academy of Pediatrics (AAP) do not recommend routine VCU in children after a first febrile UTI.[9] The AAP guidelines, which are meant for children aged 2 to 24 months old, suggest limiting the use of VCU to children identified as having complex UTIs or that have *any* abnormalities on ultrasound. We recommend beginning with an ultrasound of the kidneys and bladder once a febrile UTI has been confirmed, proceeding to evaluation with VCU if the ultrasound is abnormal or after a second febrile infection (Fig. 62.4). Other factors that may be considered as indications for VCU after the first febrile UTI include atypical causative pathogen, complex clinical course, known kidney scarring, or other congenital anomalies of kidney and the urinary tract.

Most children diagnosed with VUR after a UTI are younger than 7 years. UTI in these patients may be associated with modifiable host

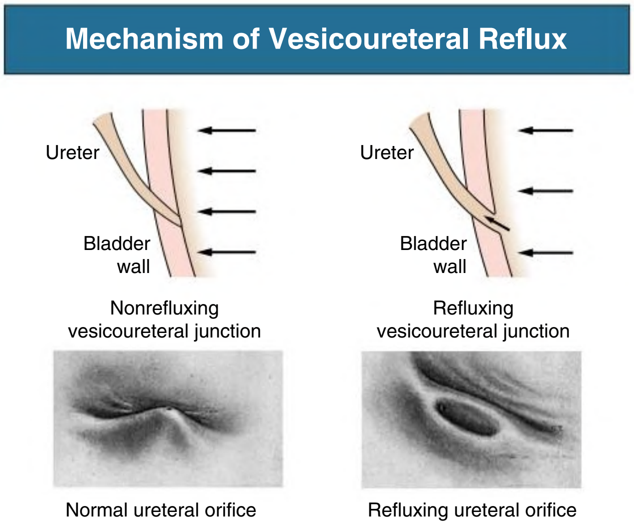

Fig. 62.3 **Pathogenesis of Vesicoureteral Reflux.** Competent *(left)* and incompetent *(right)* vesicoureteral junctions and ureteral orifices.

BOX 62.1 Clinical Presentations of Vesicoureteral Reflux

- Complicated urinary tract infection: usually acute pyelonephritis in infants and children
- Asymptomatic
  - Detected by fetal ultrasound
  - Detected in the workup of members of an affected family
  - Detected in pregnant persons with urinary tract infection
  - Detected during workup of kidney stones in children
  - Detected during assessment of other urologic congenital abnormalities

factors, such as bladder and bowel dysfunction (BBD). BBD is typically identified in toilet-trained children with urinary urgency, frequency, and/or incontinence in conjunction with constipation and is felt to be secondary to long-standing urine holding patterns. BBD is associated with development of UTIs that may be potentiated by the presence of VUR. Toilet-trained children with VUR identified after a UTI have a 43% incidence of dysfunctional voiding.[10]

Vesicoureteral Reflux in a Sibling

Approximately one-third of siblings of an index patient with VUR also have VUR.[5] There is a slightly higher incidence of VUR in female siblings of female index patients; 75% of children with VUR identified by evaluating siblings are asymptomatic, and siblings diagnosed with reflux have a better prognosis compared with the index patient with VUR, with lower risk of infection and long-term kidney injury.[5,11] The incidence of kidney damage is also lower in the siblings diagnosed with reflux compared with the index patient with VUR.[11] UTI with progression of scar was noted in only 5% of siblings with VUR observed for 3 to 7 years, and most of those with grades I and II VUR had spontaneous resolution.[11] The more benign course of sibling reflux compared with reflux identified after a UTI has led many to suggest limiting the testing of siblings. Currently, no routine testing for VUR is advised for siblings of a child with known VUR unless the sibling develops a UTI.

Other Presentations

An increased risk for kidney calculi has been reported in children with VUR. Recurrent infections with urease-splitting organisms can lead to staghorn calculi. VUR or RN may also be discovered in adults after recurrent lower or upper UTI; indeed, about 5% of sexually active females with UTI have VUR. VUR may first present as an infection in a pregnant person. In pregnant persons with active reflux, the risk for pyelonephritis is increased. A meta-analysis of 434 females with 879 pregnancies with RN showed that there was an increased risk of pregnancy-induced hypertension and pulmonary embolism.[12] Thus, it is likely that kidney scarring, not VUR itself, poses the greatest risk of morbidity in pregnant persons.[13]

TABLE 62.2 Prevalence of VUR in Patients With Urinary Tract Infection, According to Age

Age	Percentage With VUR
2–3 days	57
3–6 days	51
2–6 months	60
7–12 months	35
1–4 years	50
5–9 years	35
10–14 years	14
14 years	10
Adult	5

VUR, Vesicoureteral reflux.

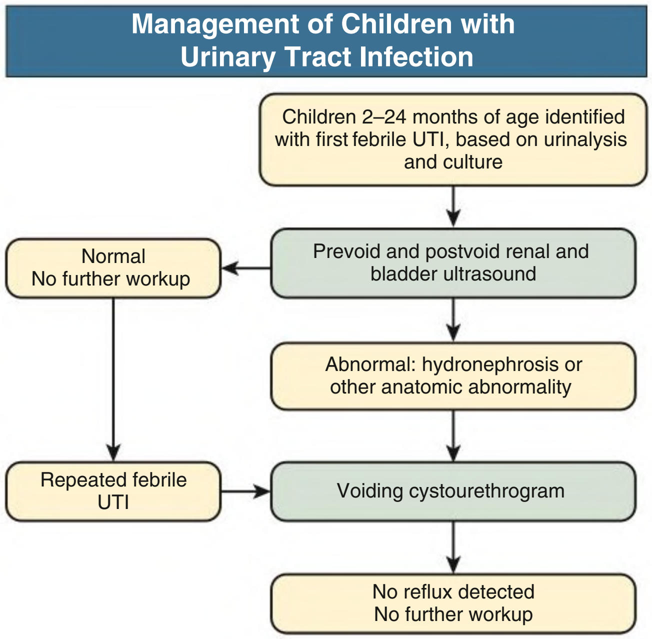

Fig. 62.4 American Academy of Pediatrics algorithm for evaluation of young children with diagnosed urinary tract infection (UTI).

Reflux Nephropathy

RN may occur in patients with VUR and can be seen in those with or without proven UTI.[14,15] RN results from one of two processes (Table 62.3).[16] Prenatal kidney injury has been postulated to be secondary to the "water-hammer" effect of high-grade reflux and occurs in the absence of infection. This typically causes kidney dysplasia. This form of kidney scarring, which is also called *congenital RN,* is more commonly noted in infants with high-grade VUR, with a greater predominance of males. The second mechanism for development of kidney injury is the combination of VUR and repeated UTI, which is also

TABLE 62.3 Types of Kidney Damage Associated With VUR

	Congenital	Acquired
Time of occurrence	Often prenatal	Postnatal, sometimes in adulthood
Previous urinary infection	Not usually	Usually
Sex	Usually males	Usually females (particularly after infancy)
Grade of VUR	Usually grades IV and V	Grades IV and V less common
Kidney scarring	Often present	Present in minority[a]
Associated bladder dysfunction	Hypercontractile bladder common	Less commonly, high-capacity bladder with incomplete voiding

[a]Depends on unilateral versus bilateral involvement and the severity of kidney involvement.
VUR, Vesicoureteral reflux.
Modified from Kenda RB, Zupancic Z, Fettich JJ, Meglic A. A follow-up study of vesico-ureteric reflux and renal scars in asymptomatic siblings of children with reflux. *Nucl Med Commun.* 1997;18:827.

BOX 62.2 Clinical Presentations of Reflux Nephropathy

- Complicated urinary infection: usually acute pyelonephritis in infants and children
- Hypertension: may be accelerated
- During pregnancy: urinary infection, hypertension, preeclampsia
- Proteinuria
- Chronic reduction in glomerular filtration rate
- Urinary calculi
- Asymptomatic
    - Detected in the workup of members of an affected family
    - Detected by fetal ultrasound
    - Detected during assessment of other urologic congenital abnormalities

called *acquired RN*. In these children, who are more commonly female, the combination of upper tract infection and reflux leads to kidney inflammation and permanent scarring. In one study, kidney cortical defects were seen on dimercaptosuccinic acid radionuclide scan (DMSA) in 45% of children with febrile UTI and VUR compared with 24% with UTI without VUR.[15] Scarring is more commonly located at the upper and lower poles of the kidney because of the anatomy of the kidney papillae in these regions.

The risk for kidney damage is greatest in the presence of high-grade VUR.[17] The Randomized Intervention for Vesicoureteral Reflux (RIVUR) trial also documented that the risk for new kidney scarring is higher in relatively older children and not in younger children as reported in the literature previously.[17] Other recent studies have also shown that older children may be at higher risk of kidney scarring.[18,19] One possible explanation for the higher incidence of kidney scarring in younger children is a lower threshold for diagnosing and investigating younger children with UTI compared with older children. Other risk factors for RN include duration of fever of longer than 72 hours before antibiotic initiation, recurrent UTI, and organisms other than *Escherichia coli*.

RN is diagnosed by using technetium-99m (^{99m}Tc)-labeled DMSA kidney scanning that demonstrates defects in the kidney outline.[20] As noted previously, high-grade prenatal reflux can lead to kidney injury in the absence of infection. Unfortunately, kidney scarring secondary to UTI is indistinguishable by DMSA kidney scan from that caused by prenatal kidney dysplasia. In addition, kidney injury can be noted after febrile UTI in the absence of identified VUR. Kidney scarring as shown by ^{99m}Tc DMSA scan correlates more closely with the severity of VUR than with a history of UTI.[21] The clinical manifestations of RN are varied and may include complicated UTI, hypertension, proteinuria, and various manifestations of CKD (Box 62.2).

The process of kidney scarring may take several years; in one study, the mean time from discovery of VUR to the appearance of a kidney scar was 6.1 years.[22] Injury is seen more at the kidney poles and is associated with clubbed calyces with medullary and cortical damage. The injury results from the local inflammatory response that may persist with chronic inflammation, tubular injury, local fibroblast activation, and interstitial collagen deposition (Fig. 62.5).[23] The loss of nephrons is associated with hyperfiltration and hypertension that can result in proteinuria and progressive loss of kidney function. This also can lead to the development of FSGS (Fig. 62.6).

Hypertension

Hypertension occurs in 10% to 30% of children and young adults with RN,[24,25] and according to one study, hypertension may take 8 years to develop from the time of diagnosis.[22] The exact cause of hypertension resulting from kidney scarring is not known, but it is thought to be caused by impaired sodium excretion resulting from the kidney injury and loss of kidney function. Hypertension is relatively uncommon in children with VUR, with an estimated probability of 2%, 6%, and 15% at 10, 15, and 21 years of age, respectively. However, hypertension increases in proportion to the degree of kidney injury.[26] Kidney scarring (noted by DMSA scans) was reported in 20% of newly diagnosed hypertension in children and adolescents.[27]

Proteinuria

Patients also may present with mild to moderate or (rarely) nephrotic-range proteinuria. Severe or nephrotic range proteinuria may suggest a histologic diagnosis of secondary FSGS, which can be confirmed by kidney biopsy if kidney size is normal and diagnosis is uncertain (see Fig. 62.6).[28] Proteinuria is commonly associated with hypertension and kidney dysfunction. CKD progression often occurs gradually over 5 to 10 years.

Chronic Kidney Disease

According to the North American Pediatric Renal Transplant Cooperative Study annual report of 2008, 3.5% of the 6491 children on dialysis had RN, which makes it the fourth most common cause of ESKD in children after FSGS; kidney aplasia, hypoplasia, or dysplasia; and obstructive uropathy.[29] The number of children with RN who present with ESKD as adults is not clear. According to one study of 123 adults with VUR diagnosed in childhood, the estimated glomerular filtration rate (eGFR) in those with nondilating VUR averaged 75 mL/1.73 m^2 and in the dilating group, the average was 72 mL/1.73 m^2; four patients (9%) in the nondilating group and 13 (17%) in the dilating group had an eGFR of less than 60 mL/1.73 m^2.[30] In a study that evaluated the clinical course of 735 children with primary VUR between 1970 to 2004, the probability of CKD was reported as 5% at

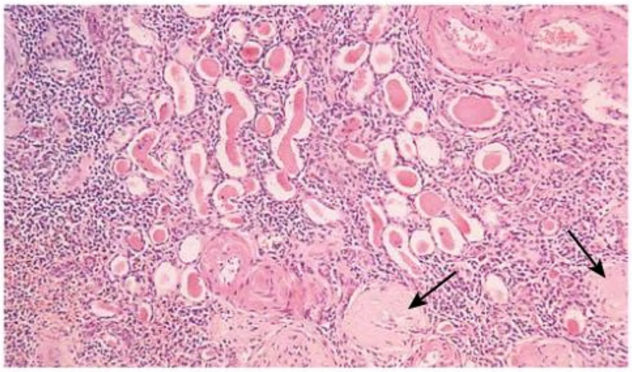

Fig. 62.5 **Histologic Changes in Reflux Nephropathy.** Sclerosed glomeruli *(arrows),* chronic inflammatory cell infiltration, and atrophic tubules with eosinophilic casts are present. (H&E stain; original magnification ×40.)

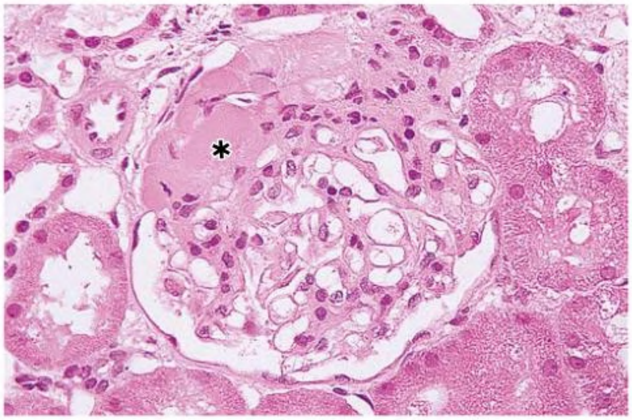

Fig. 62.6 **Focal Segmental Glomerulosclerosis (FSGS) in Reflux Nephropathy.** Light microscopy of a glomerulus from a patient with reflux nephropathy shows FSGS with the scarred area marked by an *asterisk.* (H&E stain; original magnification ×400.)

10 years after diagnosis of VUR. For children diagnosed after 1990, the probability of CKD was only 2%, indicating that the prognosis for patients diagnosed with VUR may be improving over time. Another study by the same group evaluated independent predictive factors of CKD in children with severe bilateral grades III to V VUR. The authors found that the probability of CKD for patients with bilateral severe reflux was 15% by 10 years after VUR diagnosis. The three variables that were independently associated with CKD were age at diagnosis of greater than 24 months, VUR grade V, and bilateral kidney damage.[26]

Diagnosis of Vesicoureteral Reflux and Reflux Nephropathy

An algorithm for diagnosis of VUR after the discovery of UTI is shown in Fig. 62.4. However, this is not followed by all practitioners, partly because of other published guidelines and the possible usefulness of antimicrobial prophylaxis in VUR, as reported by the RIVUR study.[31] An example of the various tests in a child with UTI and VUR is shown in Fig. 62.7.

Kidney Ultrasound

Ultrasound is the initial modality for the evaluation of postnatal hydronephrosis and UTI in children. Ultrasound is also used in siblings of children with VUR to determine whether kidney dilation suggestive of high-grade reflux is present. Although ultrasound can suggest the possibility of high-grade VUR, it is not sensitive for diagnosis of acute pyelonephritis. In patients with acute pyelonephritis, abnormalities compatible with the diagnosis were reported in 20% to 69% by ultrasound compared with 40% to 92% by DMSA scintigraphy.[32]

Kidney ultrasound is not diagnostic for VUR and is not a sensitive method for diagnosis of kidney scars. Nonetheless, a kidney ultrasound is useful to diagnose congenital anomalies of kidney and urinary tract and to detect complications of acute pyelonephritis such as kidney abscess or pyonephrosis.

Voiding Cystourethrography

VCU is the primary diagnostic modality for identification of VUR. It requires catheterization. The grading of VUR is based on radiographic appearance by VCU (see Fig. 62.1). In children with UTI, VCU should be performed as soon as the child has completed antibiotic therapy.

The results of VCU can be affected by size, type, and position of the catheter; rate of bladder filling; height of the column of contrast media; state of hydration of the patient; and volume, temperature, and concentration of the contrast medium. These parameters are not currently measured in the grading system. Durability of grade also appears to be hampered by the variability of a radiologist interpretation of VCU images. In an analysis comparing the interpretation of two blinded RIVUR radiologists with a local radiologist's interpretation, all three radiologists agreed on VUR grade in only 59% of ureters.[2] Ureteral dilation may be present without calyceal dilation, leading to discrepancies with grading.[33]

Nuclear cystography has been used to reduce the radiation exposure for children during follow-up of VUR. Nuclear cystography, although more sensitive, does not permit specific grading of VUR or reveal other anatomic defects, such as ureterocele and diverticulum. Therefore, nuclear cystography is typically not the primary study performed for identification of VUR, but it is useful in determining improvement or resolution of reflux during follow-up or after surgical correction.

DMSA Kidney Scintigraphy

DMSA scintigraphy is currently the gold standard for diagnosis of acute pyelonephritis and kidney scarring with a high rate of sensitivity. Single-photon emission computed tomography (SPECT) DMSA scintigraphy is superior to planar imaging for detection of kidney cortical damage.[34,35] The sensitivity of DMSA scintigraphy in experimentally induced acute pyelonephritis in a pig model was reported to be 92% when correlated with histologic findings.[35] By use of standardized criteria for its interpretation, high levels of intraobserver and interobserver agreement were reported.[36,37]

An abnormal DMSA scan during a febrile UTI allows for the identification of children with kidney inflammation who are at risk for development of kidney scars. For acute pyelonephritis, DMSA scintigraphy can be performed within 2 to 4 weeks after the onset of UTI symptoms; however, it is not encouraged because it usually does not change clinical management. Some have suggested the use of DMSA scanning as an initial modality to identify children with abnormalities that may suggest the need for evaluation for VUR. DMSA scintigraphy to identify kidney scarring should be performed 6 months after acute infection to allow reversible lesions to resolve.[38]

Dysplasia secondary to congenital reflux will appear similarly to kidney scarring after postnatal infections. In a child presenting with VUR, obtaining a baseline DMSA kidney scan allows identification of kidney dysplasia and scarring, which can then be observed over time.

Magnetic Resonance Imaging

Magnetic resonance imaging (MRI) can be used for the diagnosis of kidney scars because it discriminates swelling from scarring, both of which would be interpreted by DMSA scintigraphy as kidney scarring.

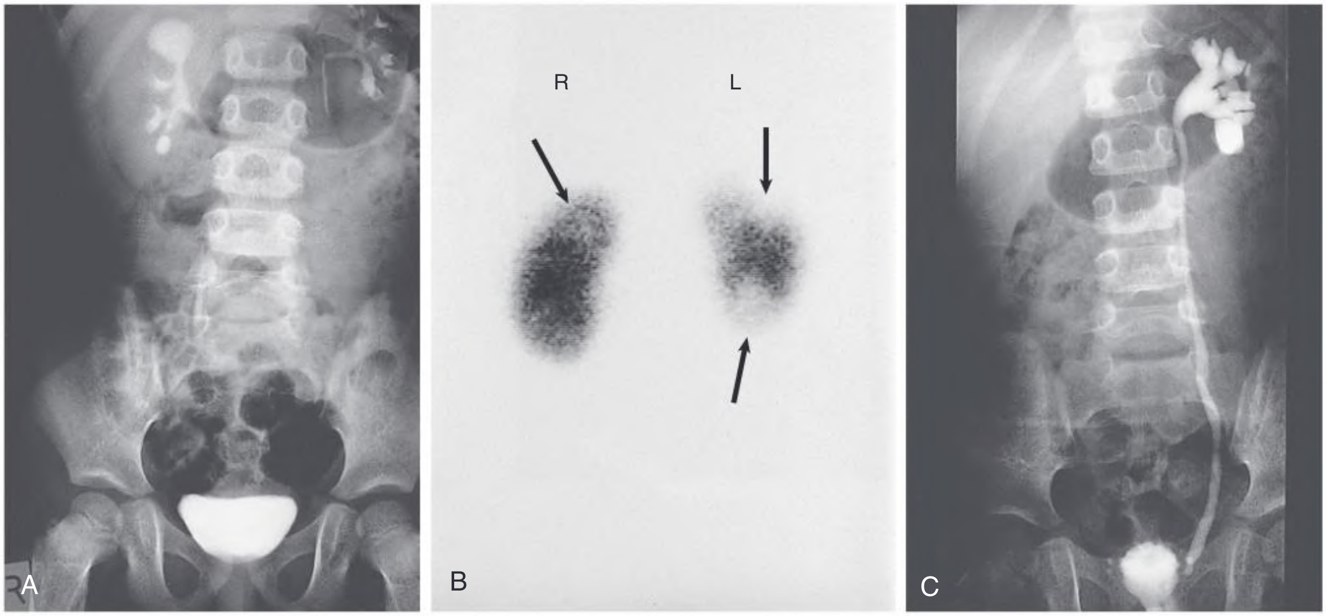

Fig. 62.7 Investigation of Reflux Nephropathy in a 3-Year-Old Child With Urinary Tract Infection. (A) Intravenous urogram showing calyceal diverticulum in the upper pole of the right kidney and kidney scarring in the upper pole and, probably, the lower pole of the left kidney. (B) Dimercaptosuccinic acid scintigraphy (posterior view) demonstrating upper and lower pole scarring *(arrows)* in the left kidney and scarring of the right upper kidney in association with the calyceal diverticulum *(arrowhead).* (C) Voiding cystourethrogram showing grade IV vesicoureteral reflux on the left.

MRI may diagnose other coexisting conditions, such as nephrolithiasis,[39] which is not diagnosed by DMSA scintigraphy. Newer imaging methods that show promise in diagnosis of kidney scarring include dynamic contrast-enhanced MRI and MRI using a gadolinium-enhanced short tau inversion recovery (STIR) sequence. However, routine use of MRI is less practical because of limited availability, especially for infants, need for prolonged sedation, and high cost. Gadolinium must be used with caution in the presence of significantly reduced glomerular filtration rate (< 30 mL/min/1.73 m^2).

Proteinuria as a Marker for Reflux Nephropathy

Proteinuria predicts CKD progression because of RN.[40] Persistent albuminuria is helpful in diagnosis of glomerular damage at an early stage. Albuminuria increases with increasing severity of VUR and kidney scarring.[41] In children with bilateral VUR with kidney scarring and normal creatinine clearance, mild albuminuria was detected in 54% of cases.[42] Evaluation for albuminuria offers the possibility of early intervention, such as the use of angiotensin-converting enzyme (ACE) inhibitors, aimed at retarding CKD progression. Severe albuminuria is usually associated with FSGS.

NATURAL HISTORY OF VESICOURETERAL REFLUX AND REFLUX NEPHROPATHY

Primary VUR, especially grades I to III, generally improves with time. This is likely the result of growth and lengthening of the submucosal segment of the ureter. Spontaneous resolution of VUR is more common with non-White race, lower grades of reflux, absence of kidney damage, and lack of voiding dysfunction. Primary VUR diagnosed before 1 year of age has a high likelihood of resolution[18,43]; however, after 1 year of age, the initial grade of reflux at presentation may be more useful to predict resolution. One retrospective study found spontaneous resolution rates to be 72% and 61% for grade I and II, compared with 32% for grade IV/V.[43] According to some studies, the resolution of VUR occurs more slowly in children with bilateral VUR compared with a unilateral VUR. The mean time until spontaneous resolution in Black children was 15 months versus 21 months in White children.[6] Increasing age at presentation and bilateral VUR decrease the probability of resolution, and bilateral grade IV or grade V VUR has a particularly low potential for spontaneous resolution.

The natural history of VUR in adults has been reported in few studies. In one study of adults (mean age of 24 years) with gross VUR diagnosed in infancy, proteinuria and CKD were present in 3 of the 13 patients with unilateral RN and 2 of the 4 patients with bilateral RN.[44] In another study of 127 adults (mean age of 41 years) with VUR diagnosed during childhood, 35% had unilateral kidney scarring, 24% had bilateral kidney scarring, 24% had albuminuria, and 11% had hypertension. Of the patients with bilateral kidney scars, 83% had reduced GFR.[45] In a study in VUR patients who were operated on, those with mild to moderate VUR and minimal kidney scarring were likely to maintain a normal GFR in adult life, whereas patients with high-grade VUR and severe scarring were likely to develop progressive CKD.[46] This suggests that the degree of kidney scarring rather than VUR grade after surgical management had the greatest correlation to progressive CKD.

Males with RN have worse outcomes compared with females. This difference has previously been attributed to a delayed diagnosis in males because of an insidious onset of proteinuria and kidney failure. In contrast, the diagnosis may be discovered earlier in females because of recurrent UTI and pregnancy-related complications. However, it is quite possible that the sex differences in adults may be because males are more likely to have congenital RN with kidney dysplasia, which may lead to greater kidney damage than acquired RN, which is seen more in females.[16]

TREATMENT

The management of VUR is based on the premise that VUR predisposes children to the development of recurrent UTI and kidney parenchymal

TABLE 62.4 **American Urologic Association Guidelines for Management of Vesicoureteral Reflux**

Grade of VUR	I	II	III	IV	V
<1 yr without UTI	Consider CAP	Consider CAP	CAP	CAP	CAP
<1 yr with UTI	CAP	CAP	CAP	CAP	CAP
>1 yr without BBD or UTI	Consider CAP	Consider CAP	Consider CAP	Possible surgery	Possible surgery
>1 yr without BBD *and* UTI	CAP/possible surgery	CAP/possible surgery	CAP/possible surgery	CAP/possible surgery	CAP/possible surgery
>1 yr with BBD *and* without UTI	Treat BBD/CAP	Treat BBD/CAP	Treat BBD/CAP/possible surgery when BBD improved	Treat BBD/CAP/possible surgery when BBD improved	Treat BBD/CAP/possible surgery when BBD improved
>1 yr with BBD *and* UTI	Treat BBD/CAP/change CAP	Treat BBD/CAP/change CAP	Treat BBD/CAP/change CAP/possible surgery when BBD improved	Treat BBD/CAP/change CAP/possible surgery when BBD improved	Treat BBD/CAP/change CAP/possible surgery when BBD improved

BBD, Bladder/bowel dysfunction; *CAP,* continuous antibiotic prophylaxis; *possible surgery,* injectable materials or reimplantation for lower grades and reimplantation for higher grades.
Modified from American Urological Association. Vesicoureteral reflux. https://www.auanet.org/guidelines-and-quality/guidelines/vesicoureteral-reflux-guideline.

injury but also has the potential for spontaneous resolution. Various treatment strategies have been used with the ultimate objective of preventing kidney injury. The two main treatment modalities are long-term antimicrobial prophylaxis and surgical correction. Surgical correction of VUR was common until antimicrobial prophylaxis for childhood UTI was introduced in 1975.[47] The International Reflux Study in Children (n = 306 patients) showed no significant difference in outcome between medical and surgical management in terms of the development of new kidney lesions or the progression of established kidney scars, although there was a lower incidence of pyelonephritis in the surgical arm.[48]

Medical Management

Medical management involves prompt antibiotic treatment of UTI, long-term antimicrobial prophylaxis, appropriate management of BBD if present, and follow-up VCU to assess the resolution of VUR and the potential development of kidney injury.

Antimicrobial Prophylaxis

The RIVUR trial revealed that trimethoprim-sulfamethoxazole (TMP/SMZ) prophylaxis reduced the risk of UTI recurrence by 50%.[31] Similar results were reported by another placebo-controlled double-blind (PRIVENT study) trial.[49] However, some other randomized studies have shown no beneficial effect with prophylaxis. Systematic reviews and meta-analyses have also reported mixed results. These variations in results have been attributed to significant differences in study designs, including patient inclusion and exclusion criteria.[50] No study has demonstrated any beneficial effect of antimicrobial prophylaxis for the prevention of kidney scarring. However, none of these studies were powered to evaluate kidney scarring as the primary outcome.

The antimicrobial agents most appropriate for prophylaxis include trimethoprim-sulfamethoxazole, trimethoprim alone, nitrofurantoin, and cephalexin. During the first 2 months of life, it is advisable to avoid TMP-SMZ because it may worsen hyperbilirubinemia and promote neurotoxicity.

Antibiotic resistance is a major risk of long-term prophylaxis and hence should be used selectively. In cases with febrile UTI and VUR, the American Urological Association (AUA) recommends continuous antibiotic prophylaxis in children less than 1 year of age and a selective approach in older children based on patient age, severity of VUR, recurrence of UTI, presence of BBD, and kidney cortical anomalies[51] (Table 62.4). The other factors that should be considered before initiating long-term antimicrobial prophylaxis include status of toilet training, risk of antibiotic resistance, anticipated compliance with daily medication administration, and the parental choice. In some cases, preemptive antimicrobial prophylaxis may be necessary to lower the risk of first UTI, such as in those with high-grade VUR diagnosed during workup for antenatal hydronephrosis.

Follow-up of patients with VUR and UTI requires rapid evaluation with stringent techniques for collection of urine specimens for culture (within 48 hours of the onset of fever) to allow early detection and prompt treatment of UTI. The timing of follow-up VCU is not well defined, but studies have suggested time intervals of 12 to 24 months.

Some studies have challenged the benefit of long-term antimicrobial prophylaxis in the prevention of recurrent infection and kidney injury in patients with VUR[52,53] and have alternatively suggested surveillance only. Concerns have also been raised regarding the potential risks for long-term antibiotic use, including the possibility of development of resistance or allergy. These studies have been used as the basis for the 2011 AAP recommendations, suggesting that evaluation for identification of VUR be limited.

The 2011 AAP guideline has affected the diagnosis and decreased surgical management of VUR by resulting in fewer VCUs ordered overall in patients 2 to 24 months old, with significant decline in VCU performed in patients with UTI.[54] The incidence of VUR has also paralleled the decrease in rate of VCU performance.[55] In a retrospective study of children age 0 to 24 months who were evaluated with both kidney ultrasound and VCU, 17% of patients in the cohort had grade IV reflux with a normal kidney ultrasound.[56] Use of the 2011 guideline may have resulted in high-grade reflux being missed in this cohort of patients.

Management of Bladder and Bowel Dysfunction

The management of bladder and bowel dysfunction may include the use of laxatives and timed frequent voiding every 2 to 3 hours. Pelvic floor exercises, behavioral modification, or anticholinergic medication have also been successfully used. A combined conservative medical and computer game–assisted pelvic floor muscle retraining decreases the incidence of breakthrough UTI and may facilitate VUR resolution in children with both BBD and VUR. Treatment of constipation by dietary measures, behavioral therapy, and laxatives helps to reduce UTI recurrence and may resolve enuresis and uninhibited bladder contractions.

TABLE 62.5 Surgical Techniques for Vesicoureteral Reflux

Technique	Success Rate (%)	Pros	Cons
Open reimplantation	95	High success rates Limited requirement for follow-up VCU Reduction in hospital stays	Surgical incision Hospitalization required Catheters needed during postoperative management Need for pain control
Endoscopic injection of dextranomer and hyaluronidase (Deflux)	70–80	Reasonable success rates Outpatient management Minimal pain	Expensive Lower success rates Need for repeated procedures Need for follow-up VCU
Laparoscopic or robotic reimplantation	70–90	Reasonable success rates Small incisions Less discomfort	Lower success rates Requires hospitalization Need for follow-up VCU Long procedure Expensive equipment Significant surgical learning curve

VCU, Voiding cystourethrography.

Surgical Management

Surgical management of VUR is now reserved for patients in whom medical management with antimicrobial prophylaxis has failed to prevent UTIs. Current indications for surgical management of VUR are recurrent infections despite compliance with a prophylactic antibiotic regimen, worsening of kidney scars as judged by DMSA scanning, and failure to comply with prophylaxis. The recent introduction of minimally invasive surgery for VUR management has made some clinicians reconsider the benefits of surgical correction as a potential first-line therapy: immediate correction could potentially avoid the need for antibiotic prophylaxis in children. Although most surgical techniques have high success rates for the correction of reflux and have been shown to reduce the risk of pyelonephritis, lower tract infections have been shown to occur in 20% of children that have had successful surgical management.[57] A review of surgical techniques is presented in Table 62.5.

Prevention of Kidney Injury

Although prevention of kidney injury remains the focus for the management of VUR, no modality has been shown to be able to consistently prevent scarring. Even the large multi-institutional studies were unable to show benefit of prophylactic regimens for prevention of scarring[17,53] because this was not a primary study endpoint and follow-up was short. Surgical ablation of VUR has also not been shown to impact the incidence of kidney scarring even with reduction in upper tract infection.

Hypertension and Proteinuria

Appropriate management of hypertension and proteinuria includes the use of ACE inhibitors or angiotensin receptor blockers (ARBs) as in other kidney diseases (see Chapter 82). Combinations of ACE inhibitors and ARBs may allow further reductions in proteinuria compared with monotherapy.[58] However, although the blood pressure control has been shown to have a beneficial effect on slowing CKD progression, the role of antiproteinuric effect in doing so is not well established. Because of the side effects of ACE inhibitors and ARBs, particularly hyperkalemia in CKD patients, we recommend monotherapy with appropriate monitoring. Combination therapy may be considered in CKD patients with poorly controlled blood pressure on monotherapy and a low risk of hyperkalemia. Historically, some patients also occasionally had their poorly functioning scarred kidney removed to help control hypertension, provided the contralateral kidney was healthy. However, this is exceptionally rare in recent times because of the availability of many potent antihypertensive agents.

SELF-ASSESSMENT QUESTIONS

1. Which imaging modality is used to diagnose vesicoureteral reflux?
   A. Ultrasound
   B. Voiding cystourethrogram
   C. DMSA kidney scan
   D. Magnetic resonance imaging
   E. Computed tomography scan
2. Which of the following patients diagnosed with VUR would have the *least* likelihood of spontaneous resolution?
   A. 2-year-old female with grade II reflux
   B. 2-year-old female with grade IV reflux
   C. 2-year-old male with grade II reflux
   D. 6-year-old female with grade IV reflux
   E. None of the above
3. Which of the following antibiotics would be an appropriate choice of prophylaxis for a 1-month-old female diagnosed with grade IV reflux?
   A. Trimethoprim-sulfamethoxazole
   B. Trimethoprim and amoxicillin
   C. Nitrofurantoin
   D. Cefalexin
   E. All of the above
4. Complications of reflux nephropathy include which of the following?
   A. Hypertension
   B. Proteinuria
   C. Progressive kidney failure
   D. Pregnancy-related complications
   E. All the above

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